

Group Forecast and Furious
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In Week 7, we selected a slightly more complex model (compared to the logistic regression from the previous week) in support vector machines (SVMs). SVMs strive to find the optimal hyperplane that separates the “up” and “down” classes with as minimal error as possible. They are able to use different kernels, which can capture non-linear relationships between variables. SVMs also tend to perform well with NLP datasets, hence the reasoning to use it on the sentiment and combined datasets in particular.

After implementing an initial support vector classifier, four hyper-parameters were tuned using f1 score as the main scoring metric. Hyper-parameters included:

- Kernel: Selects the shape of the decision boundary. One of [linear, rbf, poly, sigmoid]
- C: A regularization parameter that determines the penalty of misclassifying observations. One of [0.1, 1, 5]
- Gamma: Determines the leverage of an individual point in the decision boundary. One of [scale, auto]
- Max Iterations: Provides the number of iterations the model is allowed to run. Leaving this too small can lead to issues of convergence. One of [1000, 2000]

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SENTIMENT DATASET

When the SVM model was evaluated on the sentiment validation dataset, the model achieved an accuracy of approximately 52.2%. The confusion matrix results indicate that the model performed somewhat better at identifying upward price movements than downward movements. Specifically, the model correctly predicted 116 “Up” observations and 15 “Down” observations, while incorrectly predicting 103 “Up” cases and 17 “Down” cases. These results suggest that although the model was able to capture some signals associated with upward price movements, it struggled to reliably distinguish between the two classes overall.

C	Gamma	Kernel	Test Accuracy	Test F1
0.1	scale	linear	0.528348	0.691396
0.1	scale	RBF	0.528348	0.691396
1	auto	linear	0.528348	0.691396
1	auto	RBF	0.528348	0.691396
10	scale	linear	0.528348	0.691396
10	scale	RBF	0.528348	0.691396
100	auto	linear	0.528348	0.691396
100	auto	RBF	0.528348	0.691396
10	auto	RBF	0.528348	0.691396
10	auto	RBF	0.528348	0.691396
100	scale	RBF	0.511064	0.562857
100	scale	linear	0.511887	0.557279
1	auto	linear	0.511887	0.557279
1	scale	RBF	0.509404	0.557279
0.1	auto	linear	0.508595	0.551282
0.1	scale	linear	0.508595	0.551282

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In an attempt to improve model performance, a grid search with 5-fold cross-validation was conducted to tune the SVM hyperparameters. The search tested combinations of the regularization parameter C (0.01, 0.1, 1, and 10), kernel types (linear and RBF), and gamma values (scale and auto), with the optimization objective being the F1-score. The best-performing model achieved a slightly higher validation accuracy of approximately 53.0%. However, further inspection revealed that the tuned model predicted all observations as “Up”, resulting in 133 correct predictions and 118 incorrect predictions. This behavior suggests that the model defaulted to predicting the majority class, likely due to class imbalance in the dataset or weak discriminative signals within the TF-IDF text features. These findings indicate that while sentiment and textual features contain some information related to stock movements, additional feature engineering, or alternative text representations such as word embeddings may be necessary to improve the model’s ability to detect both upward and downward price movements.

HISTORICAL PRICE DATASET

For the historical price dataset, the hyperparameters evaluated are C (0.1,1,10,100), Gamma (scale, auto), and Kernel (linear, RBF). The following are the results of a 5 fold cross-validation grid search using these hyperparameters:

C	Gamma	Kernel	Test Accuracy	Test F1	Train Accuracy	Train F1
0.1	scale	linear	0.4856	0.4860	0.5208	0.5208
0.1	auto	linear	0.4856	0.4860	0.5208	0.5208
1	auto	linear	0.4856	0.4860	0.5208	0.5208
1	scale	linear	0.4856	0.4860	0.5208	0.5208
10	auto	linear	0.4856	0.4860	0.5208	0.5208
10	scale	linear	0.4856	0.4860	0.5208	0.5208
100	scale	linear	0.4856	0.4860	0.5208	0.5208
100	auto	linear	0.4856	0.4860	0.5208	0.5208
10	auto	RBF	0.5127	0.4801	0.5497	0.5212
10	scale	RBF	0.5127	0.4801	0.5497	0.5212
100	scale	RBF	0.5127	0.4801	0.5497	0.5212
100	auto	RBF	0.5127	0.4801	0.5497	0.5212
1	auto	RBF	0.5102	0.4318	0.5446	0.4725
1	scale	RBF	0.5102	0.4318	0.5446	0.4725
0.1	scale	RBF	0.4881	0.3487	0.5306	0.4011
0.1	auto	RBF	0.4881	0.3487	0.5306	0.4011

The results indicate that despite tuning hyperparameters, the SVM model's performance does not improve significantly. The RBF kernel did slightly outperform the linear kernel, which suggests the relationship between the predictor variables and stock price direction is nonlinear.

The performance metrics also show that the SVM did not perform well at predicting stock price directions in the historical price dataset. This, combined with the results from the logistic regression model last week, suggests that the features in the historical price data are weak at predicting stock price direction. While there is still another model that needs to be assessed, it is unlikely that the dataset creates strong predictions.

COMBINED DATASET

When using a 5-fold cross validation grid search across these hyper-parameters, f1 score was used as the primary scoring metric while accuracy was used as a secondary scoring metric. The following results were obtained:

C	Gamma	Kernel	Max Iterations	Training Accuracy	Training F1	Test Accuracy	Test F1
0.1	Scale	Linear	1000	100%	1.00	100%	1.00
0.1	Auto	Linear	1000	100%	1.00	100%	1.00
5	Auto	Poly	1000	100%	1.00	100%	1.00
5	Auto	Linear	1000	100%	1.00	100%	1.00
1	Auto	Linear	1000	100%	1.00	100%	1.00
1	Auto	Poly	1000	100%	1.00	100%	1.00
1	Scale	Linear	1000	100%	1.00	100%	1.00
1	Scale	Poly	1000	100%	1.00	100%	1.00
5	Scale	Poly	1000	100%	1.00	100%	1.00

I did not have the space to include all hyper-parameters, but there were some clear patterns between top and bottom performing models. First, the max_iterations hyper-parameter did not matter at all, performance was identical between 1000 and 2000. Second, the kernel mattered significantly and the linear and polynomial kernels outperformed the RBF and sigmoid decision boundary shapes. The regularization parameter affected performance mildly, and C=1 appeared to be consistently the best value. Gamma did not appear to matter significantly, scale and auto values both produced quality performance.

There were many grid searches that did perfectly on each individual cross-section. When combined, the model looked to do perfectly across the entire dataset. However, when tested across the entire validation dataset at once, the top model (the first row in the table above) achieved a 55% accuracy and .69 F1 score on the “up” direction class. This is closer to the expected performance, and is a step above the performance of the logistic regression (51% accuracy).